

Personal Statement

How can today's technology help families who live apart feel truly present to one another? And what would it take to build digital twins of people, objects, and scenes that not only look realistic but also behave correctly under touch, motion, and control? Inspired by these questions, my research focuses on computer graphics and vision with an emphasis on physically based simulation. By making high-fidelity digital twins easier to create and dependable in use, I aim to bring presence, learning, and care closer to everyday life. With these goals, I am applying to the PhD program in Computer Science at the USC Viterbi School of Engineering.

Finding My Research Interests. I spent real time asking myself two simple questions that many others asked me as well: "What do you want to do?" and "What are your interests?" I first met these questions years ago during my undergraduate study at the University of Toronto, where majors are chosen after the first year. I explored widely: economics, statistics, and astronomy before realizing that mathematics and computer science best match how I think and what I want to build. That dual training still anchors me today: math gives me structure in optimization, numerics, geometry, and CS gives me the systems instincts to turn ideas into tools.

When I entered graduate study, I made a choice to narrow my focus and bring rigorous code together with aesthetics and interaction through graphics and vision. During my master's at University of Southern California, I pursued that curiosity deliberately across computer graphics: CSCI 420 Computer Graphics, CSCI 520 Computer Animation and Simulation, and CSCI 580 3D Graphics and Rendering. I didn't just enjoy these courses; I excelled in them — finishing at the very top of the class — and, more importantly, learned how theoretical choices such as discretizations, time integration, conditioning surface as practical tradeoffs in real code. That experience clarified my research identity: I want to develop robust, high-fidelity physically based simulation methods for deformable and articulated objects, so that virtual worlds behave as convincingly as they look.

Research in Computer Graphics. My research experiences at UCLA's AIVC Lab made this direction concrete. I worked on 3D object generation with articulated parts and saw a clear gap between visual realism and practical use. Many systems stop at surfaces and textures, yet artists and downstream applications need models whose parts are controllable and whose articulation can be expressed as valid URDF rather than arbitrary segmentation. I led the end-to-end pipeline. I began with disciplined surveying and triage, reading closely when a paper seemed near our idea, decomposing its pipeline, mapping each step against ours, and separating surface resemblance from true overlap. I then moved quickly from thought experiments to working prototypes, authoring minimal URDFs from scratch, transferring URDFs across related objects to test articulation assumptions, and scaling from toy scripts to a full diffusion-transformer pipeline. I experimented with lightweight adaptation such as LoRA and structure-aware conditioning in the spirit of ControlNet, and I defined measurable criteria for success: no self-intersection at joints under motion, consistent joint limits under actuation, stable contact, and meshes and textures that do not destabilize downstream use. Because I was trained as a software engineer during a year-long co-op, I wrote modular, documented, and tested code so others could reproduce and extend our results. This phase taught me that visual plausibility only matters when the asset also behaves correctly under physics.

As a lighthearted aside, our project had a plot twist. We initially set out to pursue 3D scene generation with articulated objects, but the literature review and hands-on replication showed that articulated-object generation itself was not yet mature enough for our goals. Many promising models were not fully open source, and several open releases produced artifacts such as broken topology, unstable rigging, and unreliable UVs that would not survive downstream use. We decided to narrow the scope deliberately: first solve robust articulated asset generation and reliable URDF translation, then return to full scene generation. That pivot taught me how research differs from course projects. There is no fixed instruction set, only open ideas that must be scoped, stress-tested, and refined from zero to one. I learned how to select a tractable entry point, design feasibility checks and ablations that reveal failure

modes early, and use those signals to steer the next iteration.

Research in Robotics. In my second year of the master's program, I joined the Robotic Embedded Systems Laboratory at USC to help design a real-to-sim pipeline for tabletop scenes. Starting from RGB-D with known intrinsics and extrinsics, we segmented objects, reconstructed meshes, estimated pose and scale, and imported assets into simulation for manipulation tasks. I still remember the lab interview: unlike earlier interviews where I mostly recited course knowledge, this time I could truly speak pipeline — we sketched failure modes and tradeoffs on the spot. Drawing on prior hands-on testing, I proposed using SAM for segmentation and Trellis for mesh reconstruction rather than building meshes from camera point clouds, which often produce artificial artifacts and low-quality surfaces. I also explained why we would not adopt several competing models at that stage and compared options such as TripoSG and Hunyuan3D, clarifying the tradeoffs for our use case. More importantly, we discussed how simulation validates grasp feasibility and how contact, friction, and restitution parameters influence downstream success. That moment convinced me research is the right long-term path. Those conversations turned simulation from a classroom topic into a decision-making tool for building robust systems.

STEM Outreach. I am committed to doing work that is useful to others. In graduate school, that means pursuing open projects that matter rather than producing results that live only on paper. I open-source my code and notes, write clear how-to guides, and maintain checklists for reproducibility and fair baselines. To date I have released seventeen projects. Each time a stranger stars a repository, I feel encouraged not for the number itself but for the signal that someone, somewhere, was able to build on what I made. My aim is simple and consistent with the vision of trustworthy digital twins: tools that other people can run, reuse, and extend.

Mentorship is the other strand of this commitment. Each year I volunteer as a mentor, I have applied to USC's Graduate Student Mentorship and Women in Engineering Mentorship programs to support incoming and junior students. I still remember the first time a mentee asked what I wished I had known when I first enrolled, it felt like speaking to my younger self. I answer candidly and specifically—how to choose courses and find a research home, how to approach faculty, how to set up reproducible environments and manage workloads—so that others can avoid avoidable detours. For me, open code and open counsel are two ways of serving the same goal: lowering the barrier for others to participate, and building a community that turns principled ideas into working systems.

Future Work. !!!!! Need to rewrite this part !!!!! My decision to pursue a PhD at USC grows directly from Professor Jernej Barbič's course in physically based simulation, which crystallized the kind of problems I want to work on and showed me how careful modeling and numerics make systems dependable. USC is where my preparation, interests, and mentorship fit best. USC is uniquely compelling because it lets me pursue physically based simulation with Professor Jernej Barbič, while also connecting to adjacent groups that strengthen perception and generation such as Professor Yue Wang and Professor Yajie Zhao. With Professor Barbič, I want to advance robust and efficient simulation that respects contact, friction, joint limits, and stability, and that can be differentiable when needed for estimation and control. My UCLA experience taught me that some goals cannot be achieved by neural networks alone without introducing artifacts; assets that appear watertight yet are physically implausible still interpenetrate or violate constraints. I am particularly interested in collision handling that guarantees non-penetration at practical time steps and in coupling deformable and articulated dynamics under real-world actuation. With Professor Yue Wang and Professor Yajie Zhao, I see opportunities to connect perception, segmentation, and three-dimensional understanding to simulation-in-the-loop pipelines so that estimated structure and materials are verified and refined by physics rather than left as purely visual outputs.

I hope to contribute to the Viterbi community beyond papers: reproducible codebases, internal tooling others can build on, reading groups that bridge graphics, vision, and robotics, and mentorship that emphasizes research hygiene and honest ablations. I also bring a perspective shaped by both analytical rigor and design sensibilities: I care about how objects look, move, and feel — from numerical stability

to the quiet correctness of a door that never clips through its frame.

Where I See Myself. I left home in middle school to study abroad, often going long stretches without seeing my family. In a traditional household where feelings are seldom spoken, distance felt larger than miles. That experience shapes my motivation: to make presence travel better, and to turn digital twins from a novelty into something that helps people feel together.

My five-year undergraduate path included a full-year co-op that taught me to ship reliable software. I enjoy coding, and I value research that others can run, reuse, and extend. I want my work to be used, not only cited. After the PhD, I hope to work in a research lab building systems that bridge simulation and perception, releasing tools the community finds as practical as they are principled. Furthering my education at USC would bring me a step closer to that goal by accelerating the science and engineering needed to make trustworthy digital twins part of everyday life.